

TROLL A OIL PLATFORM, Norway

WMO: 010877

Lat: 60.633N

Lon: 3.717E

Elev: 295

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.3	32.0	12.6	10.6	33.3	17.4	13.4	34.9	72.5	37.1	68.1	37.8	25.1	150	1.128

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	68.3	59.8	64.6	57.7	62.4	57.4	61.3	66.6	59.8	63.3	58.5	61.3	14.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	76.2	61.9	58.6	74.1	61.3	57.1	70.4	60.2	27.4	66.0	26.3	62.9	25.5	60.6	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
61.2	54.7	49.5	DB	27.3	76.8	4.8	4.9	23.8	80.3	21.0	83.1	18.3	85.8	14.8	89.4
			WB	23.9	63.3	4.6	3.2	20.6	65.6	17.9	67.5	15.3	69.3	12.0	71.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	40.0	39.6	40.5	43.6	47.9	52.0	56.6	57.7	54.3	49.7	45.4	42.5
	DBStd	7.34	3.62	3.68	3.83	4.17	4.87	3.34	4.77	3.01	3.36	3.28	3.77	3.83
	HDD50	1627	309	290	294	199	99	15	0	0	4	42	143	232
	HDD65	6386	774	710	759	643	532	389	271	228	320	474	589	697
	CDD50	726	0	0	0	6	32	75	204	238	134	33	4	0
	CDD65	12	0	0	0	0	1	0	10	1	0	0	0	0
	CDH74	23	0	0	0	0	1	0	22	0	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
	WSAvg	24.8	29.8	30.2	24.8	21.7	22.2	21.1	19.7	20.6	23.0	26.9	26.8	31.5
	PrecAvg	73.1	8.0	6.7	6.2	4.3	3.3	3.9	4.4	5.6	6.5	7.8	7.8	8.8
(15) Precipitation	PrecMax	96.5	14.3	13.5	12.6	8.4	5.1	8.7	6.9	10.0	11.4	14.1	14.4	15.7
	PrecMin	51.8	1.5	0.6	1.3	1.1	0.8	0.5	1.0	2.2	2.2	2.4	1.8	2.2
	PrecStd	11.3	3.4	3.1	2.7	1.5	1.1	1.8	1.4	1.8	2.9	2.7	3.3	3.5
	PrecStd	11.3	3.4	3.1	2.7	1.5	1.1	1.8	1.4	1.8	2.9	2.7	3.3	3.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	46.8	48.6	58.9	68.3	64.8	77.1	69.6	64.8	58.7	54.0	50.4
		MCWB	47.6	45.8	45.4	47.3	56.5	54.8	62.3	58.4	57.0	52.4	51.5	49.3
	2%	DB	46.8	46.4	46.8	54.0	62.4	60.6	71.5	66.0	62.3	55.8	52.2	49.7
		MCWB	45.9	45.4	45.2	45.9	53.5	54.9	62.0	58.3	55.9	52.2	49.7	48.8
	5%	DB	46.5	45.0	46.4	50.4	57.5	57.4	66.5	63.0	60.6	55.1	51.7	48.4
		MCWB	45.5	43.9	45.0	44.8	50.4	53.8	59.8	58.3	55.7	51.6	49.1	47.4
	10%	DB	46.0	44.6	44.9	48.2	53.7	55.6	62.5	61.2	58.9	53.7	50.2	47.9
		MCWB	44.8	43.5	43.3	44.9	49.8	52.7	58.6	57.9	55.4	50.5	48.2	46.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.6	46.7	47.1	49.2	57.6	57.6	65.7	62.1	59.2	55.7	52.7	50.0
		MCDB	47.7	46.7	47.3	52.3	67.2	61.6	70.7	65.0	62.1	56.7	53.7	50.3
	2%	WB	46.5	45.2	46.2	47.6	54.2	55.8	62.6	60.3	57.6	54.0	51.5	48.5
		MCDB	46.7	45.4	46.4	50.3	60.5	58.7	70.7	63.2	60.1	55.0	52.5	48.7
	5%	WB	45.6	44.1	45.0	46.5	52.1	54.6	60.6	59.6	56.9	52.9	50.2	47.5
		MCDB	45.9	44.6	45.3	48.6	56.0	57.1	66.0	62.1	59.3	54.2	50.9	47.8
	10%	WB	44.3	43.3	43.9	45.4	50.1	53.4	59.0	58.6	56.0	51.6	48.9	46.3
		MCDB	45.0	44.0	44.5	47.2	52.7	55.7	61.9	61.0	58.4	53.3	49.7	46.9
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	4.8	5.0	5.3	6.0	4.9	5.3	5.1	4.4	4.3	4.3	5.0
		MCDBR	5.3	5.3	5.5	8.9	12.9	8.7	12.2	7.8	6.8	5.4	4.5	5.1
		MCWBR	6.1	5.9	5.2	4.8	6.0	5.1	5.7	5.0	4.4	5.0	5.1	6.3
	5% WB	MCDBR	5.0	4.9	4.6	6.4	11.1	7.6	10.0	6.5	5.6	5.0	4.4	5.2
		MCWBR	6.3	6.1	5.6	4.6	6.1	5.2	5.5	4.9	4.6	6.1	5.6	6.8
(40) Clear-Sky Solar Irradiance	taub		0.317	0.343	0.358	0.378	0.364	0.359	0.375	0.371	0.361	0.340	0.312	0.398
	taud		2.103	2.237	2.311	2.311	2.427	2.476	2.452	2.459	2.449	2.349	2.169	2.443
	Ebn at Noon		125	199	241	259	273	277	268	260	240	200	127	83
	Edh at Noon		14	22	28	34	32	31	31	29	25	19	13	10
(44) All-Sky Solar Radiation	RadAvg		66	232	594	1151	1624	1682	1520	1174	714	339	103	42
	RadStd		9	39	82	117	118	182	121	98	64	44	14	4

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	N/A	+1.17	N/A	N/A	N/A	N/A	-149	-212	N/A	N/A

Nomenclature: See separate page